



ENERGY
TRANSITION
PARTNERSHIP

Powering Prosperity and Enabling Sustainability in South East Asia

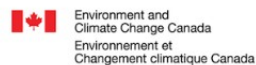


GEOHERMAL ENERGY

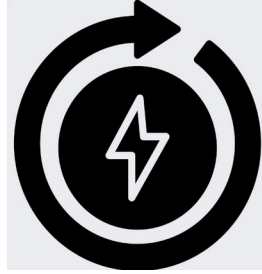
Introduction to Geothermal Reservoir and Drilling Engineering

Programme 2 – Level 1

Delivered by:



Module Overview



RE Advancement



Strategic Leadership	Tactical Management	Technical Operational
GESI Neutral	GESI Accommodative	GESI Responsive

- **Geothermal Reservoir**
 - Geothermal System
 - Heat and Mass Underground
- **Drilling Engineering**
 - Drilling Phases
 - Objectives
 - Type of Wells
 - Major Activities
 - Drilling Risks & Their Mitigation

Learning Outcomes & Criteria

Learning Outcomes

- Understand geothermal project planning & management

Criteria

- Ability to describe geothermal reservoirs and describe drilling engineering principles for effective exploration.

01

Geothermal Reservoir

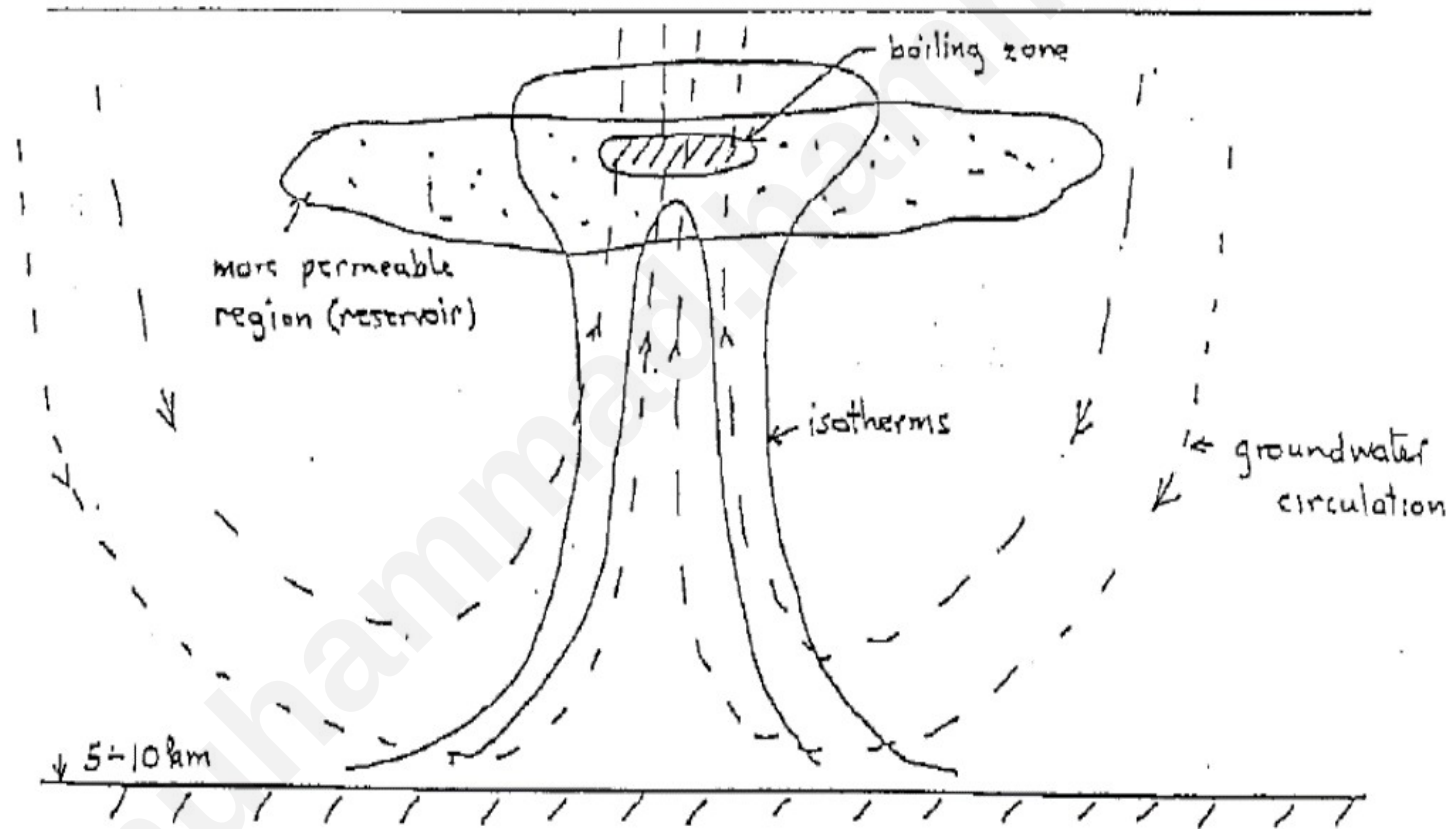
Types of Geothermal Systems

- At geothermal areas the heat flow is much larger than the world average of 60mW/m^2
- The type of geothermal system depends on the heat flow and the geological setting (discussed earlier)
- They range from warm water systems where there is no fluid movement and no boiling up to convection two-phase systems with lots of boiling underground

Approximate Thermodynamic Classification of Geothermal Systems

Category		Temperature (T)	Production Enthalpy (h)
Warm water (low temperature)		$T < 130^{\circ}\text{C}$	$h < 550 \text{ kJ/kg}$
Hot water (intermediate temperature)		$T < 230^{\circ}\text{C}$	$h < 990 \text{ kJ/kg}$
Two-phase (high temperature)	low-enthalpy	$230^{\circ}\text{C} < T < 270^{\circ}\text{C}$	$990 \text{ kJ/kg} < h < 1100 \text{ kJ/kg}$
	high enthalpy	$230^{\circ}\text{C} < T < 330^{\circ}\text{C}$	$1500 \text{ kJ/kg} < h < 2000 \text{ kJ/kg}$
	vapour - dominated	$230^{\circ}\text{C} < T < 330^{\circ}\text{C}$	$2600 \text{ kJ/kg} < h < 2800 \text{ kJ/kg}$

Convective Geothermal System



Heat Transfer Mechanisms In Geothermal Systems

- **Conduction:**

Heat flow = conductivity x temperature gradient

- **Convection:**

Heat flow = mass flow x enthalpy

- **Counter-flow:**

Heat flow = steam rising – water trickling down

Heat Transfer in Geothermal Systems

Category		Heat transfer mechanisms
Warm water (low temperature)		Conduction
Hot water (intermediate temperature)		Convection
Two-phase (high temperature)	low-enthalpy	Strong convection Some counter-flow
	high enthalpy	Moderate convection Counter-flow
	vapour - dominated	Negligible convection Counter-flow, conduction

Geothermal Reservoir

- A Geothermal Reservoir Is Quite Different From An Oil Or Gas Reservoir, Or Even A Ground Water Reservoir.
- First, A Geothermal Reservoir Is Usually Not A Clearly Defined Highly Permeable Region Confined By Low Permeability Strata;
- Secondly The Quantity To Be Extracted, Namely Heat Energy, Is Not Contained Entirely In The Reservoir Fluid.
- In An Oil Reservoir, Once The Oil Has Been Extracted The Reservoir Is Exhausted, But In A Geothermal Reservoir The Water (Or Steam) Which Was Originally In The Reservoir Can Be Replaced By Surrounding Cooler Water Which Is Heated By The Reservoir Rock And Then Becomes Available For Production.
- This Process Of **Recharge** Is Very Important In The Behaviour Of Geothermal Reservoirs

02

Drilling Engineering

Phases of Drilling

Why are we drilling?

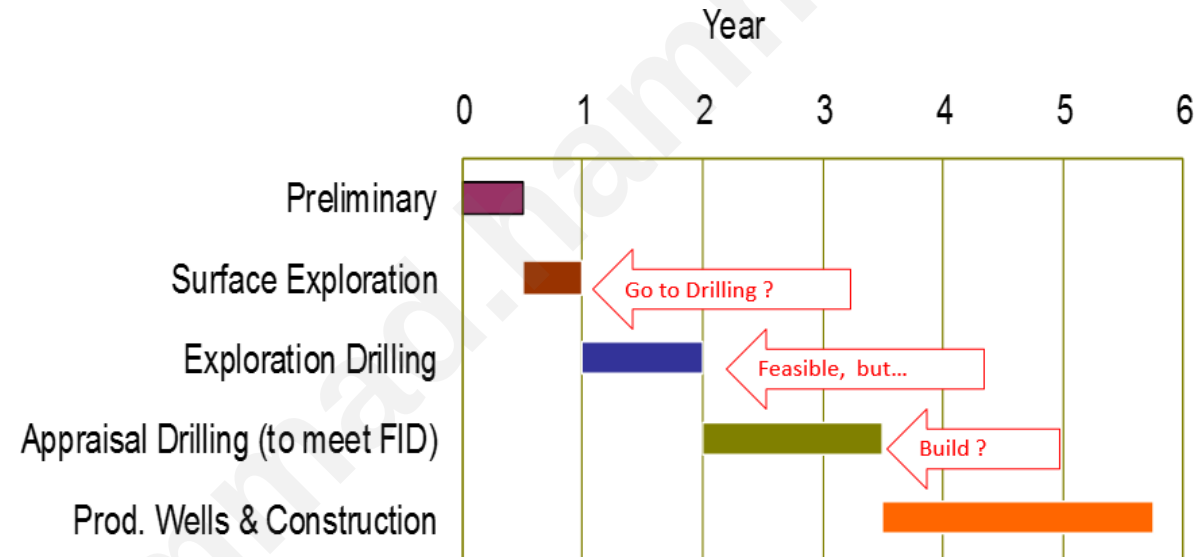
What are the
objectives?

The answer to these questions depend on what Phase
drilling you are in....



Phases of Drilling

- Exploration
- Delineation
- Appraisal
- Development
- Make-up



Each of these Phases have different objectives...

Exploration Drilling

Objective:

To prove the existence of a geothermal resource.

Simultaneously, we seek to gather as much information as possible on the reservoir; temperature, pressure, permeability geology, chemistry...

- Start by drilling the 'sweet spot'...

The wells drilled can be either slim / core holes or standard sized wells.

Well success:

Success is defined by achieving positive results from the well(s), finding a geothermal resource and indications that it is capable of development.



Delineation Drilling

Objective: (sub group of exploration)

To seek the boundaries of a geothermal field in order to gain understanding of the resource / reservoir size / MW capacity SUFFICIENT to support the intended project size (capacity that the market can accept, could range from 10MW to 300MW)

Well success:

Success is determined if the extent of the resource is defined.

In this instance, even a 'cold' well can be considered a 'success' if it identifies the edge of a resource.

Appraisal Drilling

Objective:

Seek to reach MW target for making a decision to proceed with the project (FID) (if required, and if not already achieved in the Exploration/Delineation)

Well success:

Drilling of wells that contribute to the total MW or injection capacity required - either production or injection capacity is deemed a success.

Development Drilling

Objective:

Seek to reach MW target of the project (COD)

Well success:

Suitable production and injection wells that will be used in production / injection strategy of a development.



Make-up Drilling

Objective:

Seek to reach MW target of the project (COD)

Well success:

Suitable production and injection wells that will be used in production / injection strategy of a development.



Differences in Drilling Phases

Well designs:

- Well size (also dependent on development size)
- Total well depths (small exploration rigs generally depth limited)
- Material selections and specifications
- Information available for detailed well design (More information becomes available the further drilling progresses with each Phase).

Rig type and capacities:

- Conventional rigs, coring rigs
- Horsepower required

Civil Infrastructure

- Water Supply system
- Pad Size, lay down area, sump size

Tests:

- Types of completion tests and samples taken during drilling.
- Exploration wells will generally take continuous core samples

Differences : Equipment & Civil Infrastructure

Item	Slimhole Drilling	Conventional Drilling
Rig size for at least 1,000 m deep well	Small truck mounted	Large (more horsepower)
Truckloads for well	6 to 10 15%	40 to 80
Production hole diameter	2.98" (NQ), or 3.78" (HQ)	8-1/2" (standard), or 12-1/4" (large)
Approximate time to complete 1,000 m	40 days	25 days
Approximate rig pad size	40 m x 40 m 20%	100 m x 70 m
Storage area	40 m x 40 m 4%	200 m x 200 m
Access road	Single lane track or helicopter	2 lane with no steep grades
Highest structure	Mast up to 15 m	Mast up to 45 m
Waste sump volume	250 cu m approximately	1,000 cu m approximately
Noise	Not a significant issue	Significant in close proximity to site
Peak water supply requirements	60 cu m per day for coring HQ holes 2%	3,500 cu m per day
Approx. Cost/meter	1,000 USD 25%	4,000 USD

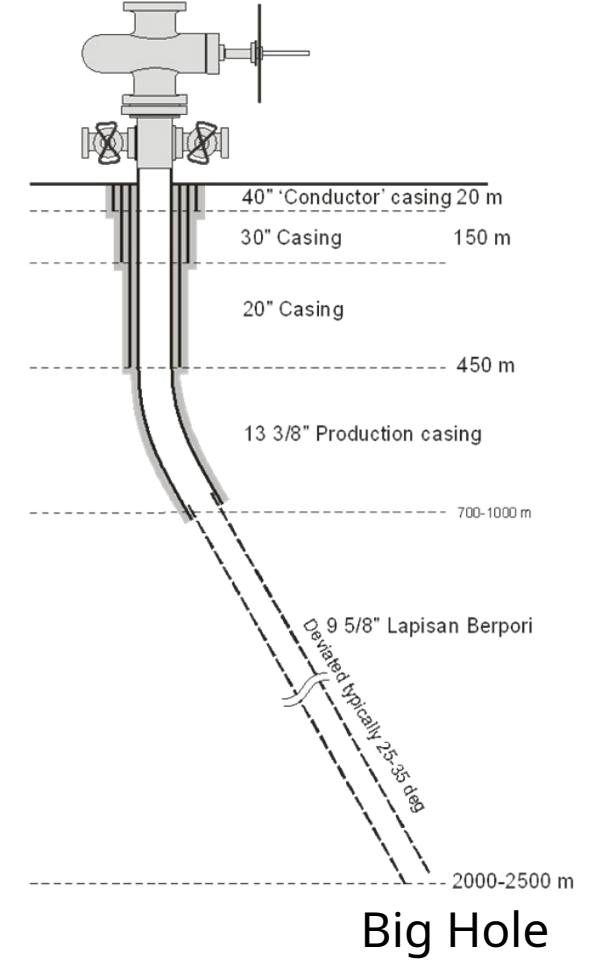
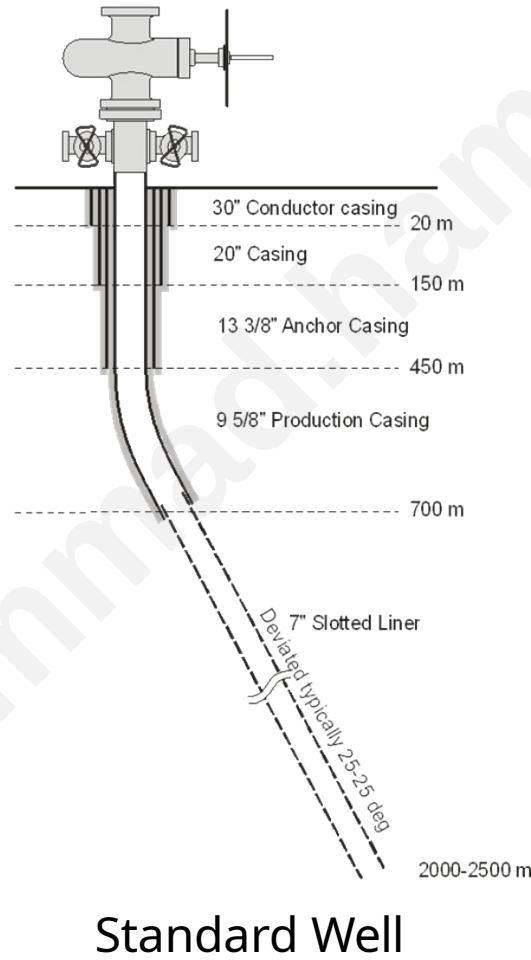
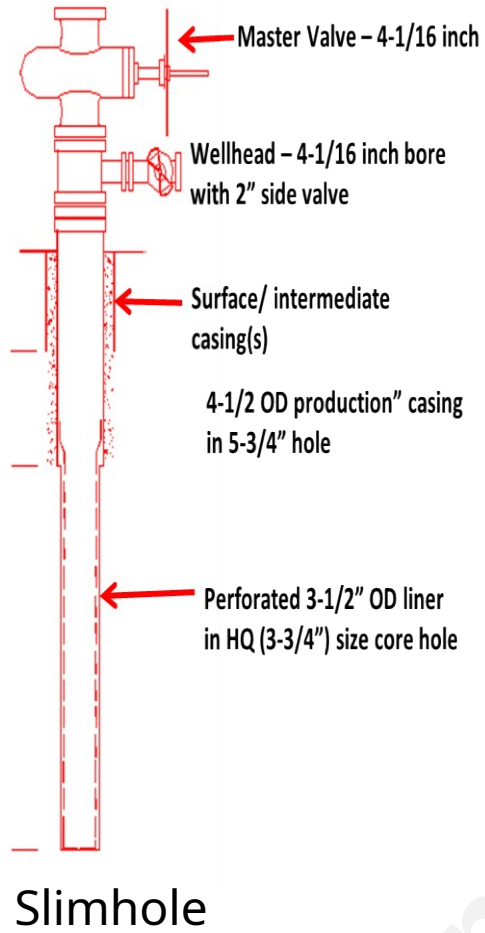


Types of Wells

There are three main types of wells employed throughout the drilling Phases...

- Slimhole -Core-hole 3-1/2" - 4-1/2" prod. casing
- Slimhole - Tricone drilled 7" prod. casing
- Standard sized well 9-5/8" prod. casing
- Big Hole well (Large diameter well) 13-3/8" prod. casing

Types of Wells



Major Activities

Many different activities are involved in drilling...

- Preparations
- Well Planning
- Inspection
- Mobilization
- Drilling / Coring
- Running in casing
- Cementing
- Directional drilling
- Drilling fluids control
- Well control
- Aerated drilling
- Coring
- Well logging
- Supervision
- Fishing



Note: “Fishing” can occur in any Phase, but we hope it is not required! This involves trying to get something out of the well that should not be there... (drill string, lost bit, broken wireline etc... hence the term ‘fishing’!!!

Major Activities : Preparation

- Pad
- Sump
- Water supply / reservoir
- Roding
- Drilling Base Camp
- Laydown and storage areas
- Cuttings pit or “zero waste”



Major Activities : Well Planning

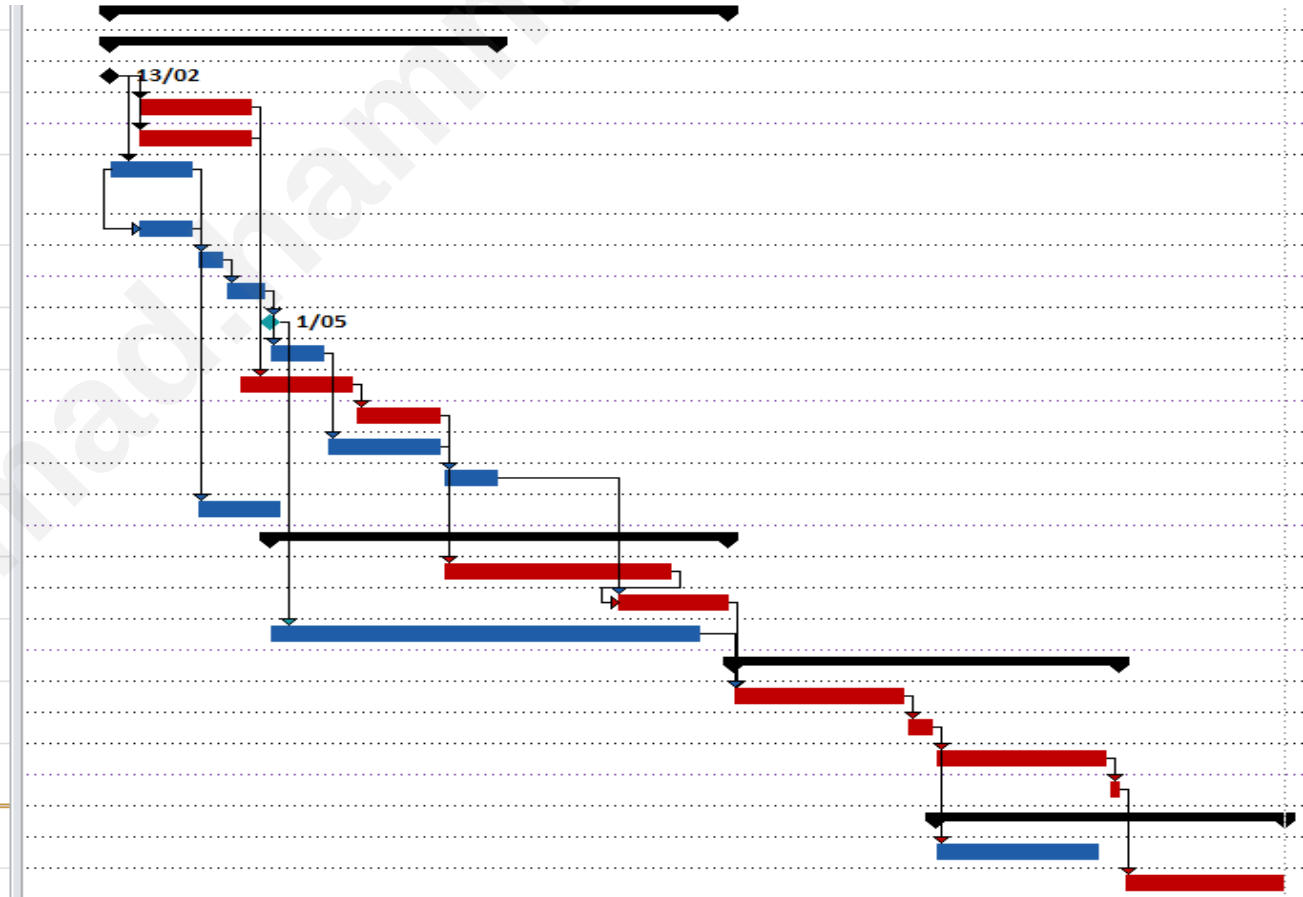
Well planning includes;

- Best possible well design as derived from the Well Delivery Process
- Followed by efficient procurement..

Well objectives	•Well objectives •Conceptual design •Geoscientific Prognosis
Detailed well design	•Primary Engineering •Material Specification
Procurement (tender and let contracts)	•Procurement plan, and tendering process •Casing and Well heads (long lead items), Drilling Rig, Materials, Services and Pad/Cellar •Well cost Estimate
Pre- Drilling	•Drilling Program •Water supply •Complete mobilisation of rig, materials and services to 1st well •Pre- spud setup
Drilling	•Supervision and Engineering •Logging and geology •Monitoring and responding to drilling variability
Testing	•Completion testing and logging •Output production testing •Water and gas chemistry

Typical Drilling Exploration Timeline

Preparation for drilling	215 days
Phase 1 - Well Planning, IPM TOR Preparation and Contractor Selection	135 days
MCG Approval to proceed	0 days
Site topographic survey	8 wks
Geotechnical survey	8 wks
Preparation of Exploration Program (including kick-off meeting, project basis, Project execution plan)	6 wks
Preparation of Project Schedule, Budget Estimate etc	4 wks
Procurement plan approval	2 wks
Tender long lead items i.e wellhead and casings	3 wks
Award Long Lead items i.e i.e wellhead and casings	0 days
Prepare tender for Drilling rig+ 10 integrated services	4 wks
Civil Engineering Design	8 wks
Civil works services tender process and award	6 wks
Integrated Services Tender process , Evaluation, Award	8 wks
Prepare Integrated Drilling services Contract	4 wks
Design of Welltest Equipment	6 wks
Site Preparation and Mobilization	160 days
Site Preparation and construction	16 wks
Rig pre-inspection and mobilization of 11 integrated services	8 wks
Delivery of long-lead wellheads and casing	30 wks
Phase 2 - Drilling	135 days
Spud and Drill First well	60 days
Rig move to second well	10 days
Spud second well	60 days
Demobilize Rig	1 wk
Phase 3 - Heat Up and testing	121 days
Heat Up and test Well 1	56 days
Heat Up and test Well 2	56 days



Major Activities : Inspections

- Independent third-party inspections are important to ensure the rig, equipment and tubulars are fit for purpose and in good condition to prevent accidents or failures.
- Rigs should be inspected by an authorised third-party prior to mobilisation to site. Any identified shortcomings need to be rectified before drilling (OES,OCS, Lloyd's Register)
- Tubulars are inspected at different frequencies during drilling to ensure they are still in adequate condition to continue drilling.



Major Activities : Inspections

Why are inspections important? For many reasons...

- Don't want the rig mast to break...
- Don't want the rig pumps to break-down...
- Don't want the generators to break down...
- Don't want tubulars to fail in the hole because the pipe wall was too thin.. or the connection threads were damaged... and so on...

This means potential saved costs (~ US\$ 45K rentals /day) on lost time and the ability to drill a well that meet objectives safely and efficiently.



Major Activities : Mobilisation

Skidding System



Well to well move in 2 days

Non-Skidding Rig



Well to well move in ~ 7 days

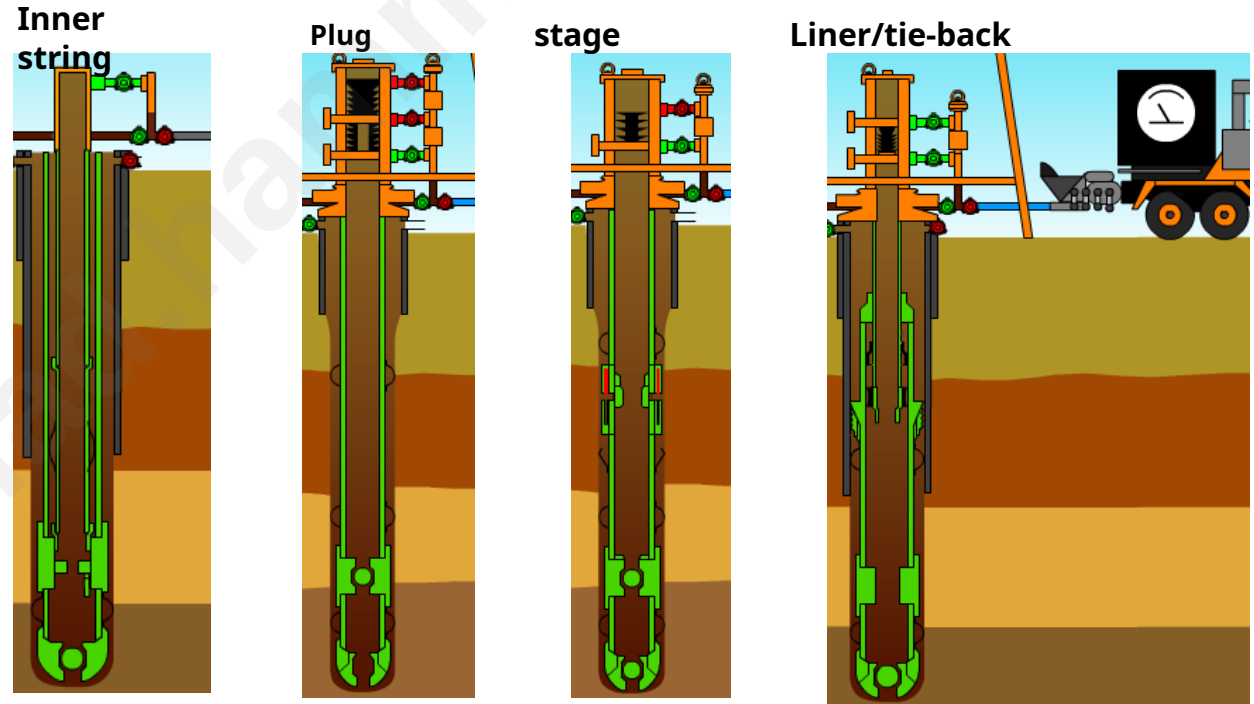
Major Activities : Running In Casing

- After reaching hole interval total depth (TD), casing is run into the hole, ready for cementing
- Centralisers placed on the outside of the casing ensure it is centered in the hole for a good cement job.



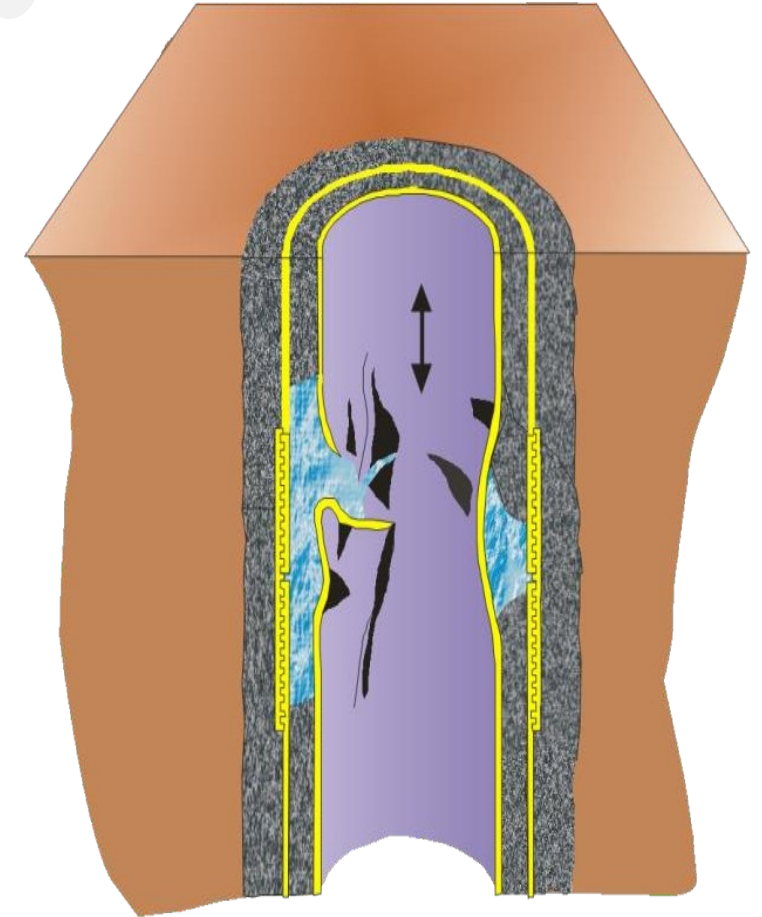
Major Activities : Cementing

- Crucial activity for long term well integrity
- Slurry designed for condition (Temperature, Pressure)
- Methods
 - Primary (inner string, plug , stage ,tied-back)
 - Secondary(remedial top job)



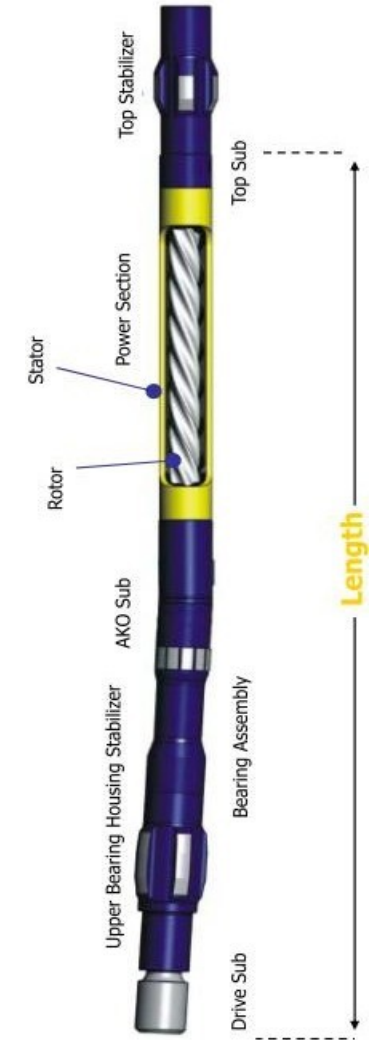
Major Activities : Cementing

- Poor cementing can lead to casing implosion or bulging; water is trapped in the annulus and expands on heat-up
- Good cement placement is critical and achieved by:
 - Casing centralization
 - Mud removal and balance plug method (no loss condition, gas migration).
 - Choice of cementing method fit for conditions



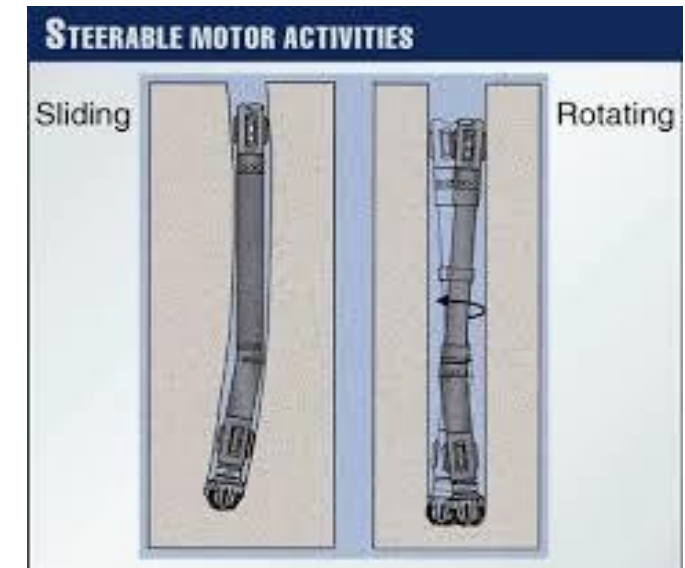
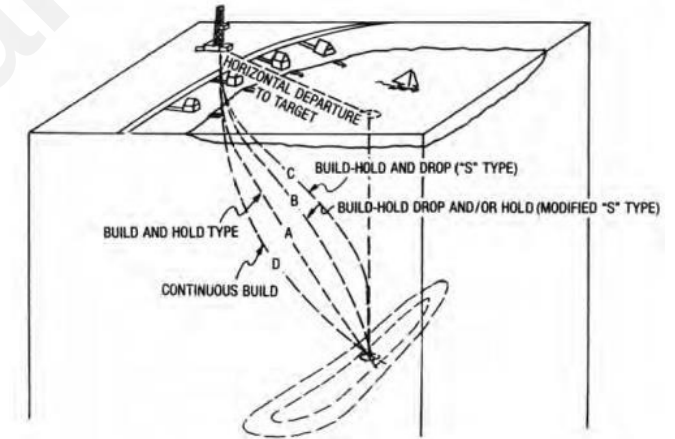
Major Activities : Directional Drilling

- Directional drilling allows us to reach sub- surface targets where it may not be possible to build a well pad directly above the target zone and drill vertically.
- Directional wells may be used to drill 'across' target faults, potentially accessing more permeability than vertical wells.
- Directional drilling allows more than one well to be drilled from a single pad.
- Mud motors are required to change direction of the well and directionally drill.
- Wells are constantly surveyed and corrected to ensure well track is drilled to plan.



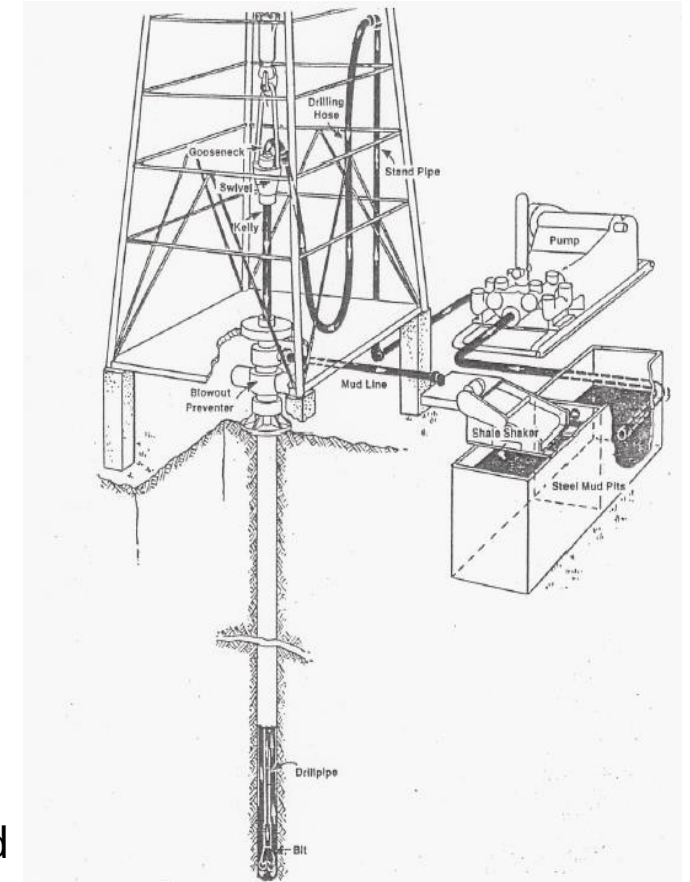
Major Activities : Directional Drilling

- Well Plan trajectory a joint effort (geoscience + well construction Team)
- Optimal design is a function of target geometry (distance from surface location and number of targets)
- Base case is a compromise between total measure depth, total depth and final hole angle
- Final hole angle not more than 45-50 to allow passage of survey tool
- Allow for 1-1/2 times the planned dogleg severity



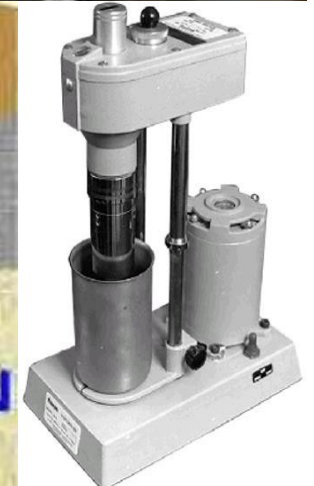
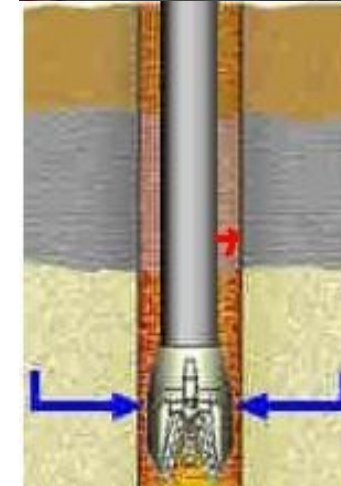
Major Activities : Drilling Fluid Control

- Drilling fluids control is required for many reasons:
 - Support wellbore
 - Cool the drill bit during drilling
 - Help bring cuttings to the surface
 - Density control (assists in well control)
 - pH control
 - Oxygen scavenger
 - Corrosion control, etc
- Various chemicals and materials are used (list next slide)
- Drilling fluids are prepared in the mud tanks, then pumped down the drill string via the mud pumps; with circulation, the fluid is then re-treated for circulation back down the hole with solids removal and dosing as required
- Use pure water or aerated water in the production hole to maximise or avoid damage to permeability



Major Activities : Drilling Fluid Control

Typical Name	Primary Application
Bentonite	Viscosifier
Barium Sulphate	Density control
Mixed Media LCM	Lost Circulation Control
Causticised Lignite	Lost Circulation Control & Deflocculant
Chrome Free Lignosulphonate	Fluid loss & rheology control
Mono Ethylene Glycol	Stuck Pipe Lubricant
High molecular weight Polymer	Viscosifier, and fluid loss control additive
Calcium Hydroxide	Flocculant
Anionic Acrylic Copolymer	High Temp Dispersant, deflocculant
Bacterially produced Polymer	Viscosifier
Sodium Hydroxide	pH control
Salts	Formation inhibition
Walnut shellsCellulose fibre	Lost Circulation Control
Sodium bicarbonate	Calcium removal
Magnesium oxideZinc oxide	pH & H ₂ S control
Tetra methyl phosphonium	Prevent bacterial decay of polymers
Sodium sulphite	Oxygen scavenger
Citric Acid	pH control
Filming Amine	Corrosion control



Well control components are crucial for safe drilling, enabling us to close-in the well if required:

-
- The image contains three photographs of industrial equipment. The leftmost image shows a large, red, cylindrical vessel with a flange top and several vertical pipes. The middle image shows a red manifold or valve assembly with multiple ports. The rightmost image shows a red skid-mounted system with several vertical cylinders and a pump, likely for gas processing.

Figure S.O.L. Land Rig Operation

Major Activities : Aerated Drilling

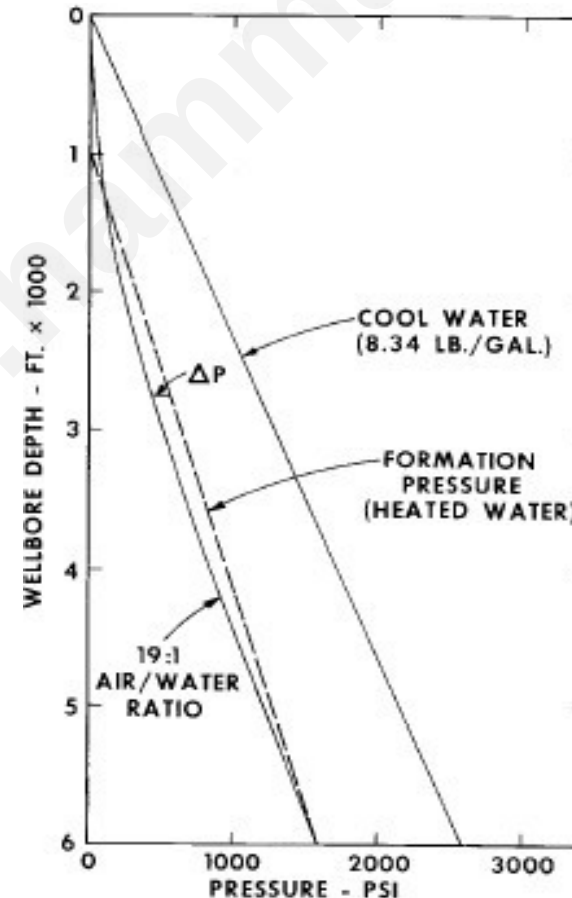
- Maintain circulation by balancing air and water flows to balance well annulus and the formation pressure
- Ability to adjust pressure balance between wellbore and reservoir during drilling

Advantages:

- Less heat-up time required for well
- Minimize well interference with adjacent wells

Disadvantages:

- Extra equipment, fuel & material costs
- High corrosion rate effect on drilling tubular



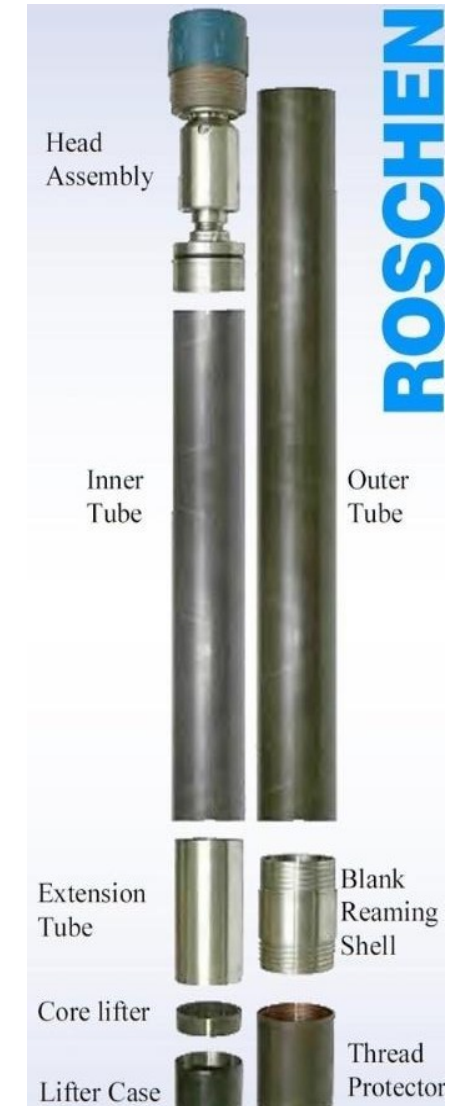
Major Activities : Coring

- Core samples are commonly taking during the Exploration Phase while coring exploration wells, usually the entire length of the well.
- Core samples provide better understanding of geology and reservoir than cuttings from rotary drilling;
 - Relationships between minerals
 - Fault and vein orientations



Major Activities : Coring

- In full size wells, spot cores can be taken intermittently as required. Cuttings samples returned to surface during rotary drilling are also retained for analysis.
- As coring is slower than rotary drilling, there is a trade-off that needs to be made between the “value of information” (gained from cores) and the “cost of information” (time/cost taken to get a core) when deciding if core samples are needed.
- Taking a core sample in a deep well might take 24 hours to trip in with a core barrel, core (6m), then trip it back out again. The cost of this operation needs to be weighed up against the value of the information it will provide.



Major Activities : Well Logging

Depending on the type of well and Phase of drilling, there are several different types of logs that can be run in a well.

- Completion tests
 - Water Loss Surveys
 - Injectivity Tests
 - Pressure Fall-off Tests (PFO)
- Pressure – Temperature – Spinner surveys (PTS)
- Heat-up Surveys
- Discharge Tests
 - Flowing PTS surveys
- Formation Imaging (FMI) logging
 - More likely to be carried out in earlier phases of drilling



Major Activities : Supervision

- Drilling supervision is key to success of the well, to ensure safe and efficient drilling to meet the objectives.
 - The drilling engineer/supervisor should run and manage the drilling operation, they are responsible for key decisions on the day to day drilling of the well.
 - Making the key decisions is crucial to risk management – who is responsible for this? The Owner or Contractor?
- Appropriate and timely decisions are crucial to efficiency. Reducing flat spots and non-productive time is key to reducing well cost.



Major Activities : Fishing

- Fishing is an unfortunate reality when something goes wrong in the drilling operation and a 'fish' is left in the hole.
- Efficient fishing operations help reduce lost time and money.
- Depending on what is lost in the hole, there are various tools available to fish it back out.
 - Magnets
 - Spears
 - Over-shots
 - Grapples



Tubular failure

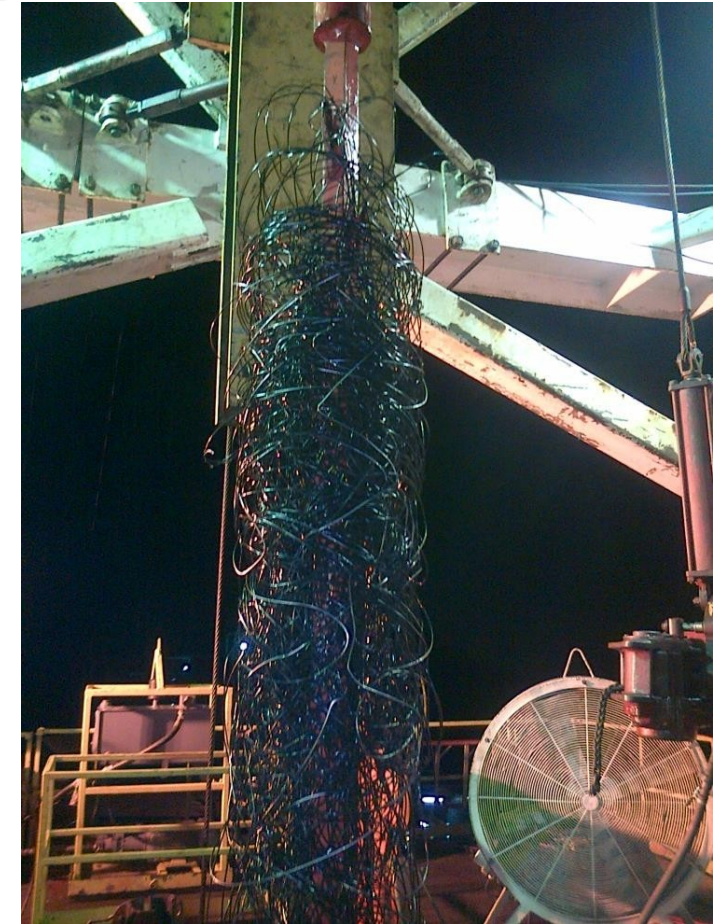


Bit failure

Major Activities : Fishing

It is best to avoid fishing by:

- good well design
- good equipment
- experienced operators, and
- good supervision



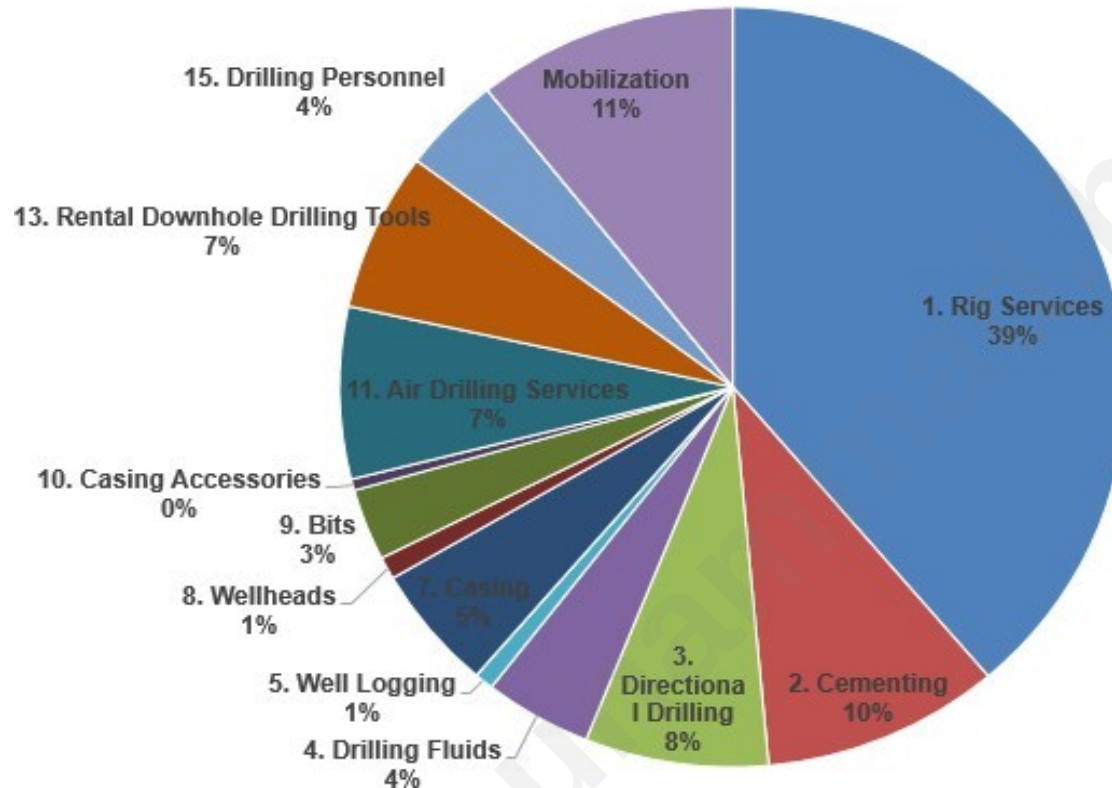
Major Activities : Others

There are many more activities involved in drilling, below is a list of some of the others...

No.	Contract
1.	Project and drilling manager
2.	Civil Works / Structural Construction/ water supply source
3.	Consolidation grouting
4.	Drilling Rig Services and catering
5.	Casing and liner (including, threading, slotting)
6.	Wellheads and Valves
7.	Cementing Services
8.	Directional Drilling Services
9.	Drilling Fluids Services
10.	Tubular Inspection Services
11.	Drill Bits
12.	Mud Logging
13.	Well Logging
14.	Well Testing Equipment
15.	Cranes, forklifts, trucks and hiabs
16.	Rental vehicles
17.	Drilling Tools and Fishing Tools Rental
18.	Back-off Services
19.	Casing Accessories
20.	Machine Shop Services
21.	Freight forwarders – custom agents
22.	Water Supply Systems operating and maintaining
23.	Fuel supply
24.	Machine shop and special welding services
25.	Drilling supervisors, rig geologists, well test engineers, and skilled Personnel

Drilling Management

Well Cost Distribution



Objective

- Drilling a well in the shortest time possible with special attention to safety and the well objectives
- Based on the premise that:
 - Time Cost Money
 - Accidents Cost Money
 - Substandard workmanship cost money

Breakdown time for a standard sized diameter well ~2,000 m MD

Drilling Risk

What are the risks in drilling?



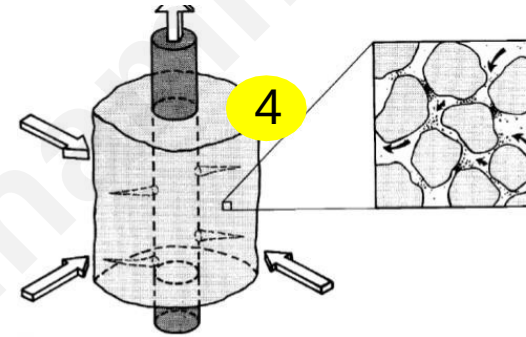
Well Blowout



Bit failure



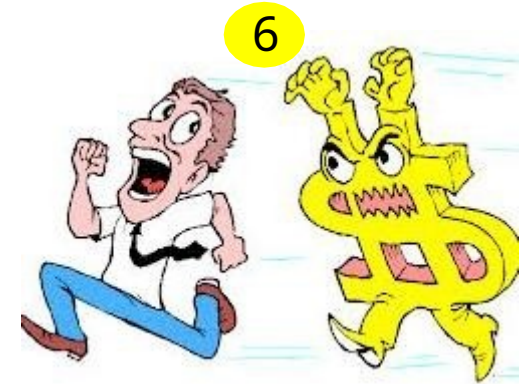
Tubular failure



Formation damage



Schedule delays



Exceed budget

Drilling Risk

Here are only some of the risks...

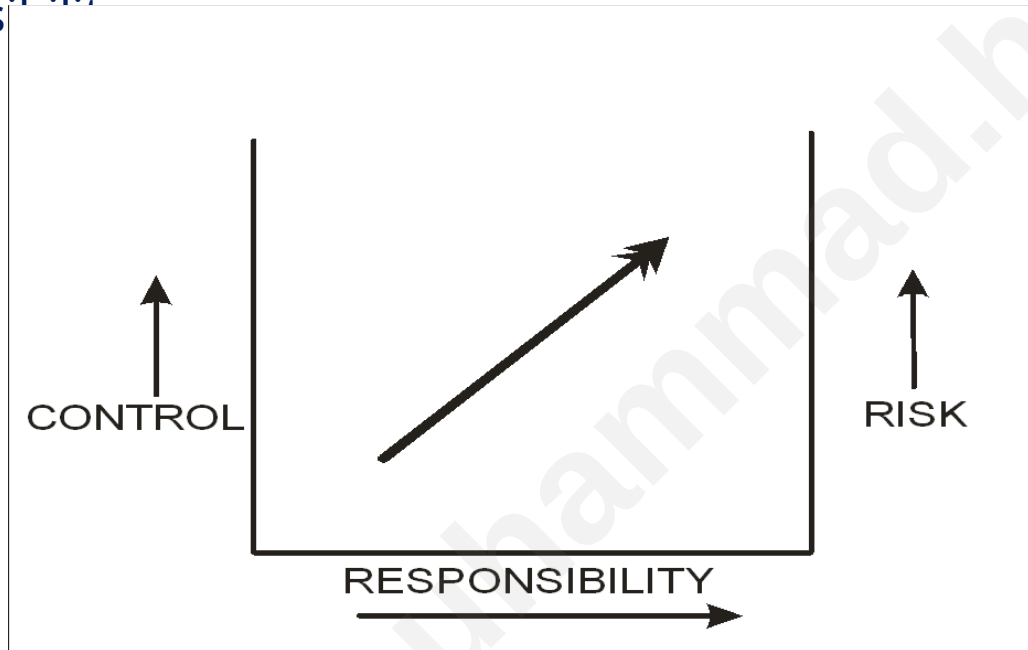
Drilling Risk	Reason
Not achieving objective	Poor prognosis, poor supervision, rig capacity, actual downhole condition
Exceeding budget	Slow drilling, stuck-in-hole, equipment failures, undersized equipment, having to side-track local opposition, excessive flat spot time and NPT etc
Schedule delays	Same as above + delays in supplies or services, poor decision tree
Unknown drilling conditions (geological risks)	Poor prognosis (risk is greatest in Exploration Phase), no offset wells
Damaging permeability	Use of bentonitic mud while drilling production hole
Losing a tool or all/part of well	Poor supervision, poor equipment, downhole condition, poor drilling practice
H2S affects employees and nearby residents	Poor understanding of geothermal fluids, inadequate safety systems and processes
Noise affects nearby residents and they stop the drilling operation	Inadequate engagement with the community

Drilling Risk

Who takes responsibility for these risks?

Owner or Contractor?

The chosen Contract and costing structure determines the responsibility.



Geothermal Well Drilling Services Contracts, United Nations University Hole. Geothermal Training Programme September 2006 Okustofnun –National Energy Authority, Iceland. Hagen Hole

Drilling Mitigation

How are the risks mitigated?

There are several ways in which drilling risk can be mitigated:

- Good planning - Well Delivery Process
- Equipment- right equipment, adequate rig capacity to achieve drilling objectives
- Drill Well on Paper (DWOP) – risk mitigation and control, Management of Change,
- Drilling Engineering and Supervision – experienced and knowledgeable
- Optimised Drilling
- Contract Structure – roles and responsibilities

Others:

- Integrated Project Team (onsite & offsite) – no silo mentality
- Good communication (RACI)
- Good reporting and recording- Lessons learnt captured and applied

Drilling Mitigation

Well objectives	• Well objectives • Conceptual design • Geoscientific Prognosis
Detailed well design	• Primary Engineering • Material Specification
Procurement (tender and let contracts)	• Procurement plan, and tendering process • Casing and Well heads (long lead items), Drilling Rig, Materials, Services and Pad/Cellar • Well cost Estimate
Pre- Drilling	• Drilling Program • Water supply • Complete mobilisation of rig, materials and services to 1st well • Pre- spud setup
Drilling	• Supervision and Engineering • Logging and geology • Monitoring and responding to drilling variability
Testing	• Completion testing and logging • Output production testing • Water and gas chemistry

Question Structure

Pre-Test Questions: Gauge basic understanding of Introduction to Geothermal Reservoir and Drilling Engineering.

Quiz Questions: Check for deeper understanding and introduce specific challenges regarding Geothermal Reservoir and Drilling Engineering.

Post-Test Questions: Ensure comprehension and critical thinking regarding Geothermal Reservoir and Drilling Engineering.

Quiz questions (2 / topic, with guided answers - to be used during topic delivery)

1. Rank the following phases of drilling in the order they typically occur in geothermal project planning:
 - Development Drilling
 - Exploration Drilling
 - Delineation Drilling
 - Appraisal Drilling
2. Word cloud: What words come to mind when you think about “geothermal reservoirs”?



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Questions and discussion

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